THE DIRECT IMAGE OF A QUASI-COHERENT MODULE ALONG A CLOSED IMMERSION IS LOCALLY A COKERNEL OF QUASI-COHERENT MODULES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The quasi-coherence of the direct image of the structure sheaf along a closed immersion of ringed spaces. Used at the start of the counted proof (source lines 2106-2107). Supplies the quasi-coherent sheaf of rings over which the presentation is written.
- The compatibility of the direct image functor with arbitrary colimits. Invoked inside the counted proof at source lines 2111-2112 as a pointer to a lemma of this chapter. Allows the local cokernel presentation to be transported from the closed subspace to the ambient space.

Source reference. Original lemma lemma-i-star-right-adjoint: sources/stacks-modules/pinned-original-source.tex, source lines 2112–2112. Navigation pointer only: the counted text refers to the Stacks item lemma-i-star-right-adjoint, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $i : (Z, \mathcal{O}_Z) \to (X, \mathcal{O}_X)$ be a closed immersion of ringed spaces. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_Z$-module. Then $i_*\mathcal{F}$ is locally on X the cokernel of a map of quasi-coherent $\mathcal{O}_X$-modules.*

Proof. This is true because $i_*\mathcal{O}_Z$ is quasi-coherent by definition. And locally on Z the sheaf $\mathcal{F}$ is a cokernel of a map between direct sums of copies of $\mathcal{O}_Z$. Moreover, any direct sum of copies of the *the same* quasi-coherent sheaf is quasi-coherent. And finally, i_* commutes with arbitrary colimits, see Lemma lemma-i-star-right-adjoint. Some details omitted. □

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